

MapReduce Applications

Recommender Systems

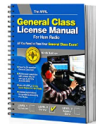


Recommender Systems

Amazon recommendations for our family

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Top picks for you

 Tetley Tea Bags, Black and Green, 72 Count (Packaging may vary) ★★★★★ 2,076 \$7.09 ✓prime FREE One-Day	 Clorox Disinfecting Wipes, Bleach Free Cleaning Wipes, Fresh... ★★★★★ 60,823 \$10.49 ✓prime FREE One-Day	 Wild Planet Wild Sardines in Extra Virgin Olive Oil, Sea Salt, Lightly... ★★★★★ 9,918 \$8.38 ✓prime FREE One-Day	 Mrs. Meyer's Gel Vinegar Cleaner, Hard Water Stain Remover and... ★★★★★ 1,172 \$5.49 ✓prime FREE One-Day	 ARRL General Class License... ★★★★★ 1,849 \$32.95 66 pts ✓prime FREE One-Day	 Charmin Ultra Gentle Toilet Paper, 18 Mega Rolls = 72 Regular Rolls ★★★★★ 39,804 \$19.99 ✓prime FREE One-Day	 Garden Tutor Dirtlizer 3-0-3 Concentrated Liquid Fertilizer &... ★★★★★ 16 \$21.95 ✓prime FREE One-Day
 Scrubbing Bubbles Toilet Bowl Cleaner and Power Stain... ★★★★★ 12,067 \$2.79 ✓prime FREE Delivery	 Amazon Basics 100 Pack AA High-Performance Alkaline Batteries,... ★★★★★ 674,775 \$26.99 ✓prime FREE One-Day	 Hosonson 100% Waterproof Vinyl PU Rectangle Heavy Duty... ★★★★★ 342 \$18.99 ✓prime FREE Delivery	 Fisher Chef's Naturals Walnut Halves & Pieces ★★★★★ 24,196 \$12.52 ✓prime FREE One-Day	 Aquafresh Extreme Clean, Whitening Action, Fluoride... ★★★★★ 13,298 \$5.08 ✓prime FREE One-Day	 Terranics Magnesium Complex, 500mg, 120 Veg Capsules... ★★★★★ 4,638 \$19.97 ✓prime FREE One-Day	 Reynolds Kitchens Quick Cut Plastic Wrap, 225 Square Feet ★★★★★ 6,060 \$3.98 ✓prime FREE One-Day



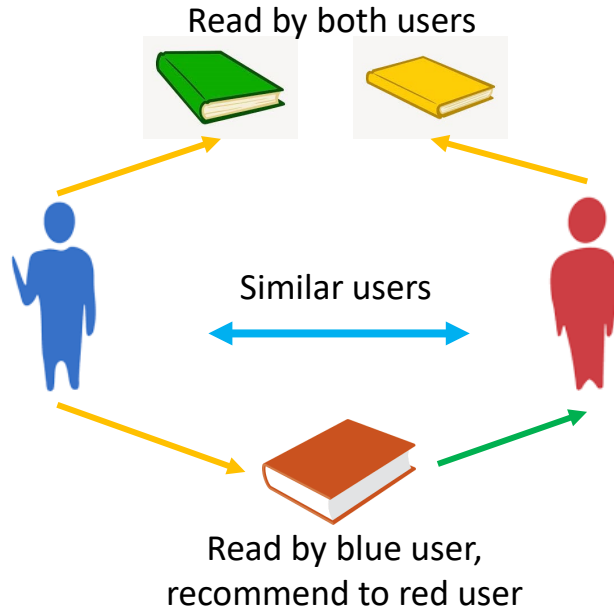
Recommender Systems – Basics

- Recommender systems have become an essential part of our daily online user experience.
- Mission critical technology in several companies.
- Success and Failure...

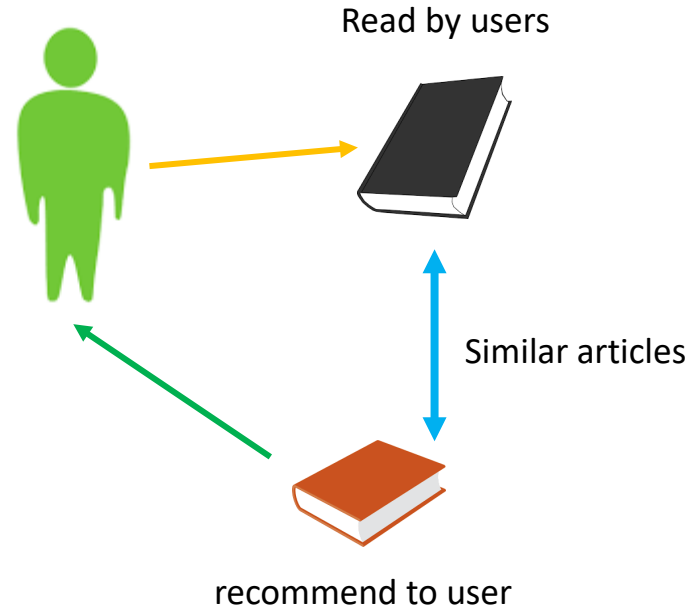


Recommender Systems – How It Works

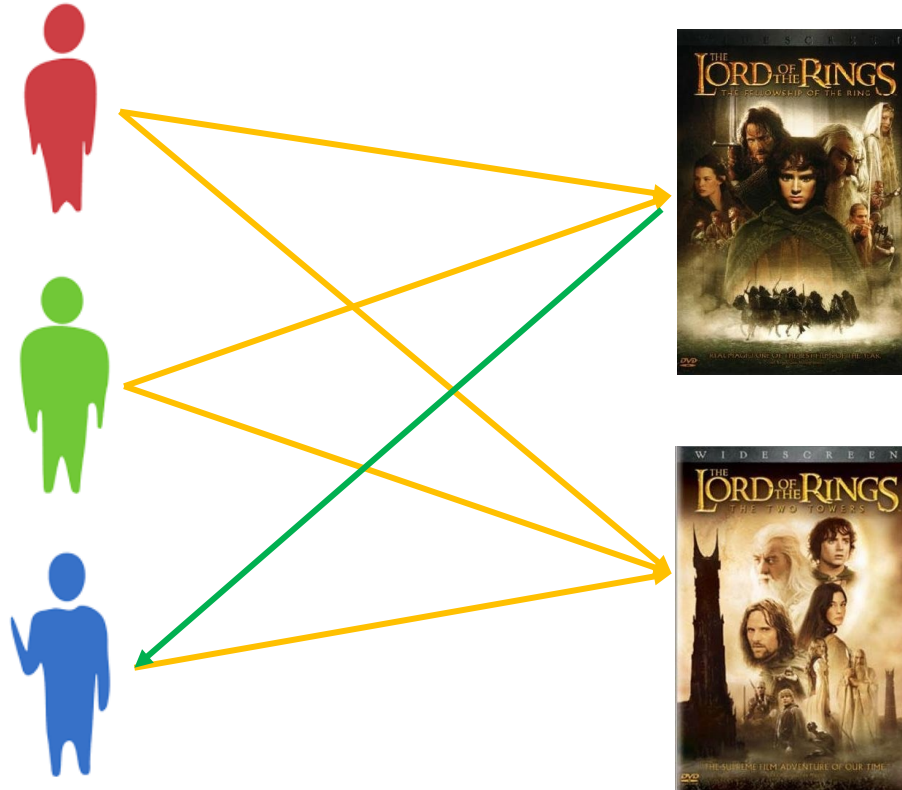
Collaborative Filtering



Content Based Filtering



Recommender System – Example



Movie Similarities

	userID ↑	movieID ↑	rating ↑	timestamp ↑
▲	A	B	C	D
1	1	1	4	964982703
2	1	3	4	964981247
3	1	6	4	964982224
4	1	47	5	964983815
5	1	50	5	964982931
6	1	70	3	964982400
7	1	101	5	964980868
8	1	110	4	964982176
9	1	151	5	964984041
10	1	157	5	964984100

	movieID ↑	title ↑	genres ↑
▲	A	B	C
1	1	Toy Story (1995)	Adventure Animation
2	2	Jumanji (1995)	Adventure Children's
3	3	Grumpier Old Men (1995)	Comedy Romance
4	4	Waiting to Exhale (1995)	Comedy Drama Romance
5	5	Father of the Bride Part II (1995)	Comedy
6	6	Heat (1995)	Action Crime Thriller
7	7	Sabrina (1995)	Comedy Romance
8	8	Tom and Huck (1995)	Adventure Children's
9	9	Sudden Death (1995)	Action
10	10	GoldenEye (1995)	Action Adventure

Objective: Identify the movie recommendations.



Movie Similarities – MapReduce Small Demo

user	movie	rating	timestamp
43	9	5	891351432
18	9	5	880130550
18	4	3	880132150
27	1	5	875975579
21	1	5	874965758
21	8	1	875072484
27	7	4	875975520
27	9	4	875975656
27	8	4	875975717
27	4	4	875981421
30	1	4	877888877
30	9	4	877889005
30	4	4	877889130
30	7	4	877892210
21	9	5	878543541
18	8	5	880130802
18	1	5	880130802
33	8	4	882140001
33	9	3	882140205
33	1	3	882140487
33	4	5	882141306
33	7	2	882396790
43	8	4	883600657
43	4	5	891351772
43	7	5	891352220



Movie Similarities – MapReduce Step 1

43	9	5	891351432
18	9	5	880130550
18	4	3	880132150
27	1	5	875975579
21	1	5	874965758
21	8	1	875072484
27	7	4	875975520
27	9	4	875975656
27	8	4	875975717
27	4	4	875981421
30	1	4	877888877
30	9	4	877889005
30	4	4	877889130
30	7	4	877892210
21	9	5	878543541
18	8	5	880130802
18	1	5	880130802
33	8	4	882140001
33	9	3	882140205
33	1	3	882140487
33	4	5	882141306
33	7	2	882396790
43	8	4	883600657
43	4	5	891351772
43	7	5	891352220

MAPPER

43 , (9, 5)
18 , (9, 5)
18 , (4, 3)
27 , (1, 5)
21 , (1, 5)
21 , (8, 1)
27 , (7, 4)
27 , (9, 4)
27 , (8, 4)
27 , (4, 4)
30 , (1, 4)
30 , (9, 4)
30 , (4, 4)
30 , (7, 4)
21 , (9, 5)
18 , (8, 5)
18 , (1, 5)
33 , (8, 4)
33 , (9, 3)
33 , (1, 3)
33 , (4, 5)
33 , (7, 2)
43 , (8, 4)
43 , (4, 5)
43 , (7, 5)

REDUCER

18, ((9, 5), (4, 3), (8, 5), (1, 5))
21, ((1, 5), (8, 1), (9, 5))
27, ((1, 5), (7, 4), (9, 4), (8, 4), (4, 4))
30, ((1, 4), (9, 4), (4, 4), (7, 4))
33, ((8, 4), (9, 3), (1, 3), (4, 5), (7, 2))
43, ((9, 5), (8, 4), (4, 5), (7, 5))



Movie Similarities – MapReduce Step 2

18, ((9, 5), (4, 3), (8, 5), (1, 5))
 21, ((1, 5), (8, 1), (9, 5))
 27, ((1, 5), (7, 4), (9, 4), (8, 4), (4, 4))
 30, ((1, 4), (9, 4), (4, 4), (7, 4))
 33, ((8, 4), (9, 3), (1, 3), (4, 5), (7, 2))
 43, ((9, 5), (8, 4), (4, 5), (7, 5))



(9, 4) , (5, 3)
 (9, 8) , (5, 5)
 (9, 1) , (5, 5)
 (4, 8) , (3, 5)
 (4, 1) , (3, 5)
 (8, 1) , (5, 5)

(1, 8) , (5, 1)
 (1, 9) , (5, 5)
 (8, 9) , (1, 5)

⋮

(9, 8) , (5, 4)
 (9, 4) , (5, 5)
 (9, 7) , (5, 5)
 (8, 4) , (4, 5)
 (8, 7) , (4, 5)
 (4, 7) , (5, 5)

(4, 9) , (3, 5)
 (8, 9) , (5, 5)
 (1, 9) , (5, 5)
 (8, 4) , (5, 3)
 (1, 4) , (5, 3)
 (1, 8) , (5, 5)

(8, 1) , (1, 5)
 (9, 1) , (5, 5)
 (9, 8) , (5, 1)

⋮

(8, 9) , (4, 5)
 (4, 9) , (5, 5)
 (7, 9) , (5, 5)
 (4, 8) , (5, 4)
 (7, 8) , (5, 4)
 (7, 4) , (5, 5)



(1 , 4) , (0.9381 , 4)
 (1 , 7) , (0.9899 , 3)
 (1 , 8) , (0.8883 , 4)
 (1 , 9) , (0.9959 , 5)
 (4 , 1) , (0.9381 , 4)
 (4 , 7) , (0.9473 , 4)
 (4 , 8) , (0.9595 , 4)
 (4 , 9) , (0.9560 , 5)
 (7 , 1) , (0.9899 , 3)
 (7 , 4) , (0.9473 , 4)
 (7 , 8) , (0.9467 , 3)
 (7 , 9) , (0.9929 , 4)
 (8 , 1) , (0.8883 , 4)
 (8 , 4) , (0.9595 , 4)
 (8 , 7) , (0.9467 , 3)
 (8 , 9) , (0.9067 , 5)
 (9 , 1) , (0.9959 , 5)
 (9 , 4) , (0.956 , 5)
 (9 , 7) , (0.9929 , 4)
 (9 , 8) , (0.9067 , 5)



Cosine Similarity

$$\text{Cosine similarity } |x, y| = \frac{xy}{\|x\| \|y\|}$$

Movie1	Movie4
5	3
5	4
4	4
3	5

$$\begin{aligned} &= \frac{5 * 3 + 5 * 4 + 4 * 4 + 3 * 5}{\left(\sqrt{5^2 + 5^2 + 4^2 + 3^2}\right) * \left(\sqrt{3^2 + 4^2 + 4^2 + 5^2}\right)} \\ &= 0.9381 \end{aligned}$$



Movie Similarities – MapReduce Step 2

18, ((9, 5), (4, 3), (8, 5), (1, 5))
 21, ((1, 5), (8, 1), (9, 5))
 27, ((1, 5), (7, 4), (9, 4), (8, 4), (4, 4))
 30, ((1, 4), (9, 4), (4, 4), (7, 4))
 33, ((8, 4), (9, 3), (1, 3), (4, 5), (7, 2))
 43, ((9, 5), (8, 4), (4, 5), (7, 5))



(9, 4) , (5, 3)
 (9, 8) , (5, 5)
 (9, 1) , (5, 5)
 (4, 8) , (3, 5)
 (4, 1) , (3, 5)
 (8, 1) , (5, 5)

(1, 8) , (5, 1)
 (1, 9) , (5, 5)
 (8, 9) , (1, 5)

⋮

(9, 8) , (5, 4)
 (9, 4) , (5, 5)
 (9, 7) , (5, 5)
 (8, 4) , (4, 5)
 (8, 7) , (4, 5)
 (4, 7) , (5, 5)

(4, 9) , (3, 5)
 (8, 9) , (5, 5)
 (1, 9) , (5, 5)
 (8, 4) , (5, 3)
 (1, 4) , (5, 3)
 (1, 8) , (5, 5)

(8, 1) , (1, 5)
 (9, 1) , (5, 5)
 (9, 8) , (5, 1)

⋮

(8, 9) , (4, 5)
 (4, 9) , (5, 5)
 (7, 9) , (5, 5)
 (4, 8) , (5, 4)
 (7, 8) , (5, 4)
 (7, 4) , (5, 5)



(1 , 4) , (0.9381 , 4)
 (1 , 7) , (0.9899 , 3)
 (1 , 8) , (0.8883 , 4)
 (1 , 9) , (0.9959 , 5)
 (4 , 1) , (0.9381 , 4)
 (4 , 7) , (0.9473 , 4)
 (4 , 8) , (0.9595 , 4)
 (4 , 9) , (0.9560 , 5)
 (7 , 1) , (0.9899 , 3)
 (7 , 4) , (0.9473 , 4)
 (7 , 8) , (0.9467 , 3)
 (7 , 9) , (0.9929 , 4)
 (8 , 1) , (0.8883 , 4)
 (8 , 4) , (0.9595 , 4)
 (8 , 7) , (0.9467 , 3)
 (8 , 9) , (0.9067 , 5)
 (9 , 1) , (0.9959 , 5)
 (9 , 4) , (0.956 , 5)
 (9 , 7) , (0.9929 , 4)
 (9 , 8) , (0.9067 , 5)



Movie Similarities – MapReduce Step 2 with Filter

18, ((9, 5), (4, 3), (8, 5), (1, 5))
 21, ((1, 5), (8, 1), (9, 5))
 27, ((1, 5), (7, 4), (9, 4), (8, 4), (4, 4))
 30, ((1, 4), (9, 4), (4, 4), (7, 4))
 33, ((8, 4), (9, 3), (1, 3), (4, 5), (7, 2))
 43, ((9, 5), (8, 4), (4, 5), (7, 5))



(9, 4) , (5, 3)
 (9, 8) , (5, 5)
 (9, 1) , (5, 5)
 (4, 8) , (3, 5)
 (4, 1) , (3, 5)
 (8, 1) , (5, 5)

(1, 8) , (5, 1)
 (1, 9) , (5, 5)
 (8, 9) , (1, 5)

⋮

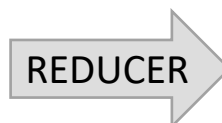
(9, 8) , (5, 4)
 (9, 4) , (5, 5)
 (9, 7) , (5, 5)
 (8, 4) , (4, 5)
 (8, 7) , (4, 5)
 (4, 7) , (5, 5)

(4, 9) , (3, 5)
 (8, 9) , (5, 5)
 (1, 9) , (5, 5)
 (8, 4) , (5, 3)
 (1, 4) , (5, 3)
 (1, 8) , (5, 5)

(8, 1) , (1, 5)
 (9, 1) , (5, 5)
 (9, 8) , (5, 1)

⋮

(8, 9) , (4, 5)
 (4, 9) , (5, 5)
 (7, 9) , (5, 5)
 (4, 8) , (5, 4)
 (7, 8) , (5, 4)
 (7, 4) , (5, 5)



(1, 7) , (0.9899, 3)
 (1, 9) , (0.9959, 5)
 (4, 8) , (0.9595, 4)
 (4, 9) , (0.9560, 5)
 (7, 1) , (0.9899, 3)
 (7, 9) , (0.9929, 4)
 (8, 4) , (0.9595, 4)
 (9, 1) , (0.9959, 5)
 (9, 4) , (0.9560, 5)
 (9, 7) , (0.9929, 4)



Movie Similarities – MapReduce Step 3 – Mapper

(1 , 7) , (0.9899 , 3)
(1 , 9) , (0.9959 , 5)
(4 , 8) , (0.9595 , 4)
(4 , 9) , (0.956 , 5)
(7 , 1) , (0.9899 , 3)
(7 , 9) , (0.9929 , 4)
(8 , 4) , (0.9595 , 4)
(9 , 1) , (0.9959 , 5)
(9 , 4) , (0.956 , 5)
(9 , 7) , (0.9929 , 4)

MAPPER

(Sabrina (1995) , 0.9899) , (Toy Story (1995) , 3)
(Sabrina (1995) , 0.9929) , (Sudden Death (1995) , 4)
(Sudden Death (1995) , 0.956) , (Waiting to Exhale (1995) , 5)
(Sudden Death (1995) , 0.9929) , (Sabrina (1995) , 4)
(Sudden Death (1995) , 0.9959) , (Toy Story (1995) , 5)
(Tom and Huck (1995) , 0.9595) , (Waiting to Exhale (1995) , 4)
(Toy Story (1995) , 0.9899) , (Sabrina (1995) , 3)
(Toy Story (1995) , 0.9959) , (Sudden Death (1995) , 5)
(Waiting to Exhale (1995) , 0.956) , (Sudden Death (1995) , 5)
(Waiting to Exhale (1995) , 0.9595) , (Tom and Huck (1995) , 4)



Movie Similarities – MapReduce Step 3 – Reducer

(Sabrina (1995) , 0.9899)	, (Toy Story (1995) , 3)
(Sabrina (1995) , 0.9929)	, (Sudden Death (1995) , 4)
(Sudden Death (1995) , 0.9560)	, (Waiting to Exhale (1995) , 5)
(Sudden Death (1995) , 0.9929)	, (Sabrina (1995) , 4)
(Sudden Death (1995) , 0.9959)	, (Toy Story (1995) , 5)
(Tom and Huck (1995) , 0.9595)	, (Waiting to Exhale (1995) , 4)
(Toy Story (1995) , 0.9899)	, (Sabrina (1995) , 3)
(Toy Story (1995) , 0.9959)	, (Sudden Death (1995) , 5)
(Waiting to Exhale (1995) , 0.9560)	, (Sudden Death (1995) , 5)
(Waiting to Exhale (1995) , 0.9595)	, (Tom and Huck (1995) , 4)

REDUCER

Sabrina (1995)	, (Toy Story (1995) , 0.9899 , 3)
Sabrina (1995)	, (Sudden Death (1995) , 0.9929 , 4)
Sudden Death (1995)	, (Waiting to Exhale (1995) , 0.9560 , 5)
Sudden Death (1995)	, (Sabrina (1995) , 0.9929 , 4)
Sudden Death (1995)	, (Toy Story (1995) , 0.9959 , 5)
Tom and Huck (1995)	, (Waiting to Exhale (1995) , 0.9595 , 4)
Toy Story (1995)	, (Sabrina (1995) , 0.9899 , 3)
Toy Story (1995)	, (Sudden Death (1995) , 0.9959 , 5)
Waiting to Exhale (1995)	, (Sudden Death (1995) , 0.9560 , 5)
Waiting to Exhale (1995)	, (Tom and Huck (1995) , 0.9595 , 4)



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